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Building Better Activation Oracles

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Abstract

Activation Oracles (AOs) are promising methods for interpreting residual stream activations. However, current AOs face important issues, such as hallucinations and vagueness. Additionally, text-inversion confounds make them hard to evaluate. To address this, we improve the Activation Oracle (AO) training regime in four ways: training on on-policy rollouts, improving the conversational dataset, feeding more layers, and an improvement to the injection formula. The capability improvements are marginal, but quality of life improvements are substantial. In addition, we open source the first comprehensive evaluation suite for AO quality, which we call **AObench**. Overall, we hope that our work sets a foundation that helps improve AOs and other models in the paradigm of scalable, end-to-end interpretability.

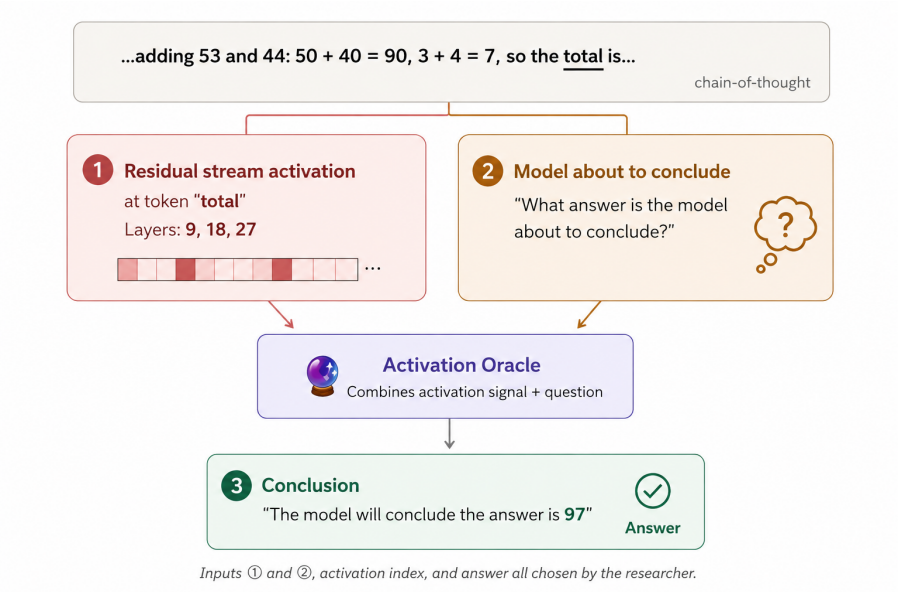


Figure 1: **Activation Oracle overview**. The Oracle receives residual-stream activations and a natural-language question, then produces an answer about the model state represented by those activations.

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1 Introduction

Activation Oracles (henceforth AOs) [Karvonen et al., 2025] are finetuned LLMs that can receive the original LLM's activations as input and answer natural language questions about them.

However, current AOs face several issues that make them hard to use, such as hallucinations, vagueness, and a lack of verifiable faithfulness [Jakkli et al., 2026]. Additionally, text-inversion confounds (where a model can match a hypothetical "true" oracle's performance by simply reconstructing the surrounding text from an activation and answer purely from this reconstructed text) make them hard to evaluate.

The standard AO training recipe comprises three components: the LatentQA conversational dataset [Pan et al., 2024], a suite of binary classification tasks, and a self-supervised past/future-lens objective trained on FineWeb, in which the AO predicts tokens before or after a sequence of activations. We identify issues with LatentQA as a dataset, and the FineWeb past/future-lens task as a training objective. Activations are fed via a norm-matched injection formula, after the second transformer layer [Karvonen et al., 2025].

We propose to partially alleviate these problems by constructing a better conversational dataset, feeding activations from multiple layers and multiple token positions, training on on-policy data and increasing the magnitude of the activation injection.

We find that these changes (particularly the new conversational dataset) produce consistent improvements: in both quantitative evaluation and qualitative testing, the resulting AO scores higher overall on evaluations, follows instructions better, hallucinates less, and is substantially less *vague* than the original AO checkpoint. To enable further work in this direction, we release **AObench**, the first comprehensive evaluation suite for AO quality, designed to measure what an ideal AO should be good at while attempting to remain robust to text-inversion confounds, all while targeting their major issues.[2]

Conversely, we find narrow post-training on the tasks from Ivanova et al. [2026] consistently fails to exceed simple linear-probe performance.

We see AOs as part of the emerging paradigm of scalable, end-to-end interpretability [Steinhardt, 2025, Pan et al., 2024, Karvonen et al., 2025, Choi et al., 2025, Huang et al., 2025, Li et al., 2026, Fraser-Taliente et al., 2026]: training models on self-supervised objectives to map model internals to natural-language explanations. We believe the largest wins for AOs will likely come from improvements to the unsupervised training task itself; our contributions take a step in this direction, but we expect substantial further headroom if a scalable task which increases capability can be found.

2 Issues with current Activation Oracles

Jakkli et al. [2026] demonstrate scenarios in which AOs are hard to work with. We focus on addressing two of their issues: **hallucinations**, where the AO outputs false information, and **vagueness**, where the AO output is generic (and therefore unfalsifiable) and does not answer the user's question.

They additionally highlight the problem of **text inversion**: the model can succeed simply by inferring the surrounding text and answering from that reconstruction, just as any black-box oracle could; this is a major frustration in evaluating AOs.

3 Improving Activation Oracle training

3.1 A better conversational dataset

To enable the Activation Oracle to answer natural language questions, a dataset consisting of questions and answers about activations is needed. To this end, the original paper used *LatentQA*. [Pan et al., 2024]

However, we found that this dataset was of low quality, likely incentivizing vagueness. We isolate three issues:

[2] Code: `https://github.com/japhba/activation_oracles`. Models and datasets: `https://huggingface.co/collections/ceselder/building-better-activation-oracles`

- The model is given a complicated prompt, and then a specific question is asked about this prompt. We think the answers to the questions LatentQA poses are often not easily retrievable from activations, which makes it a difficult task for the AO, not incentivizing much beyond text inversion, and may even directly incentivize hallucinations/guessing if the relevant info is not present.
- The questions are not about on-policy data, but about specifics of a user prompt: this does not target the model's internal reasoning.
- It was generated by o1, a now outdated model.

We constructed a new conversational dataset that attempts to address all of these concerns. Because we don't want the questions to be trivially answerable from adjacent tokens (text inversion), we construct QA pairs as follows: a separate LLM (Sonnet 4.6) is given the target model's chain-of-thought (CoT), and is instructed to split the chain-of-thought into a prefix and suffix, and to write a question about the suffix. It is instructed to do this in a way such that the question is hard to answer purely from the text of the prefix (i.e. to avoid text inversion), but plausibly answerable from the prefix's activations (solvability). [3]

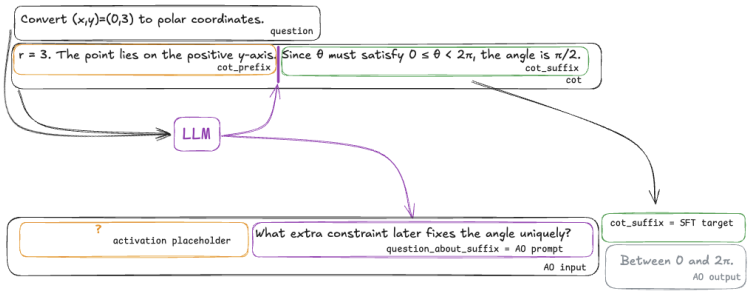


Figure 2: **How our conversational dataset is constructed.** A language model is asked to split a chain-of-thought and to produce a question about the split suffix that is plausibly answerable from the split prefix.

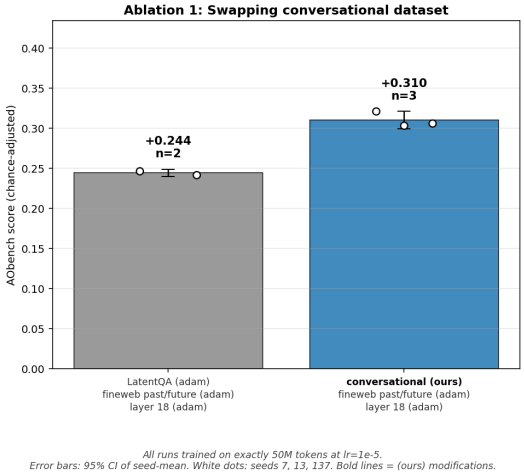


Figure 3: **Conversational dataset swap, isolated.** Replacing only LatentQA [Pan et al., 2024] with our conversational dataset (leaving past/future-lens corpus and layer choice fixed) improves chance-adjusted AObench score from $+0.244$ to $+0.310$ ($n = 3$ seeds). This is the single largest step in our recipe.

[3] You can explore our dataset here: `https://huggingface.co/datasets/ceselder/cot-oracle-convqa-chunked-sonnet`

We ablate the effect of this task in Fig. 3 by replacing only LatentQA in Adam's recipe (leaving everything else the same) and notice a significant uplift, across the board on our AObench evaluations. We find that the responses are more specific and the resulting model is less vague, and responds better to instructions.

3.2 Layer choice/feeding multiple layers to the AO

Adam originally fed activations randomly selected from either layer 25%, 50% or 75% of total model depth. Since most features live around the 55-80% layer ranges, we suspected a layer sweep could be important. Indeed, we find that AO performance peaks at layer 22 (62%). Feeding 5 contiguous layers from layer 21-25 causes further uplift. Interestingly, the largest uplift is on model diffing tasks. We'd like to point out that training a multi-layer Activation Oracle can cause an increase in training time due to longer context, and that most gains can be had by simply choosing a layer at 65% depth (though this may differ per model, and per application).

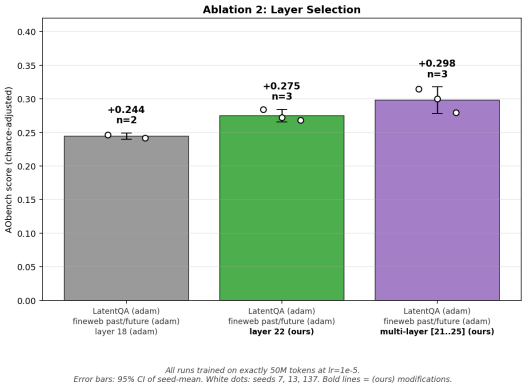


Figure 4: **Layer sweep.** Layer 22 causes improved performance over Layer 18 (+0.025 on AObench), and 5 contiguous layers even further still (+0.05 on AObench)

3.3 Training on on-policy data

To train Activation Oracles, we need scalable unsupervised training tasks. A common way to achieve this is to predict past and/or future tokens from the activations, known as past or future-lens. This requires some data to source activations from, from which then to predict tokens.

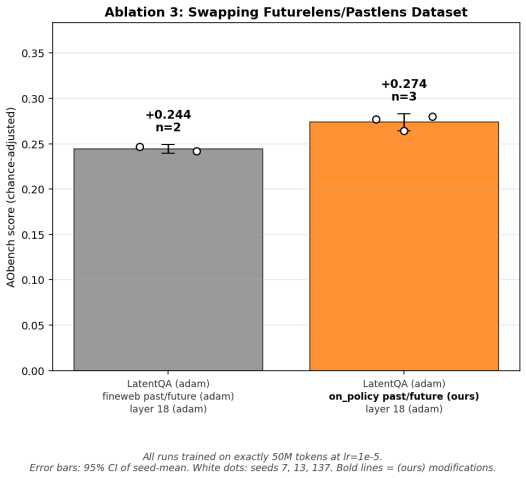


Figure 5: **On-policy data.** Replacing only the past/future-lens corpus from FineWeb to on-policy chain-of-thought rollouts improves chance-adjusted AObench score from $+0.244$ to $+0.274$ ($n = 3$ seeds), a smaller effect than the conversational swap.